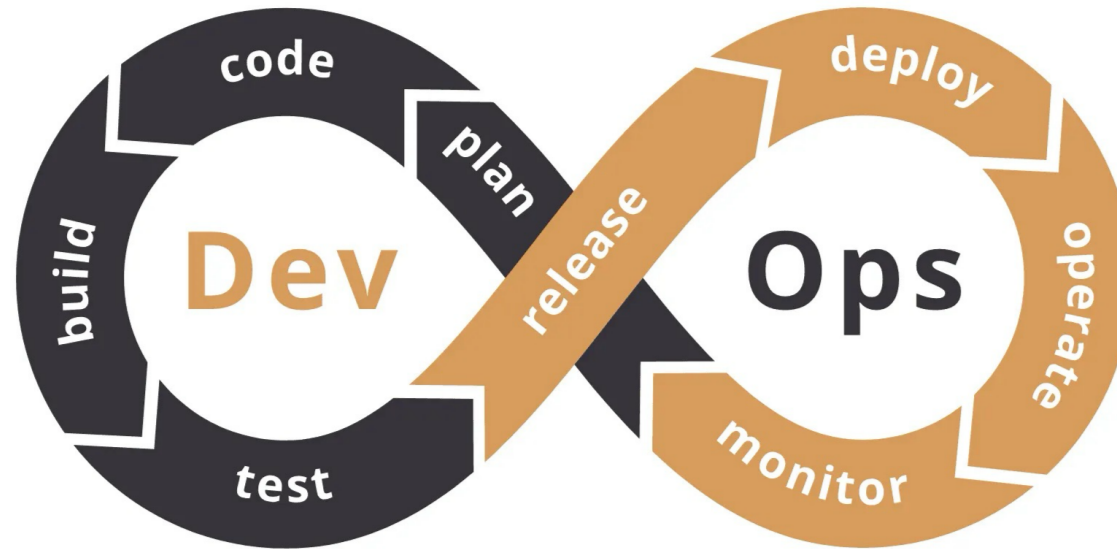




Trainer: Nilesh Ghule

Software Development Methodologies

Module Contents

- ① Software Engg
 - Core concepts
 - SDLC
 - Agile dev model
 - Scrum
 - Atlassian Jira
 - Software Testing
 - Selenium

- ② DevOps
 - SCM: GIT
 - Containerization
 - Docker
 - Orchestration
 - Docker Swarm
 - Kubernetes
 - CI/CD pipeline
 - Jenkins

③ Cloud Computing

DevOps

- ① Development
 - Developers & Testers
 - ↳ Programming, ...
 - App Testing, ...
 - Sw Engg - SDLC
 - Agile - Scrum
- ② Operations
 - Deployments
 - CI/CD mgmt
 - Monitoring

Module Evaluation

- ① CCEE: 25 m
- ② Lab: 20 m
- ③ Internals: 20 m

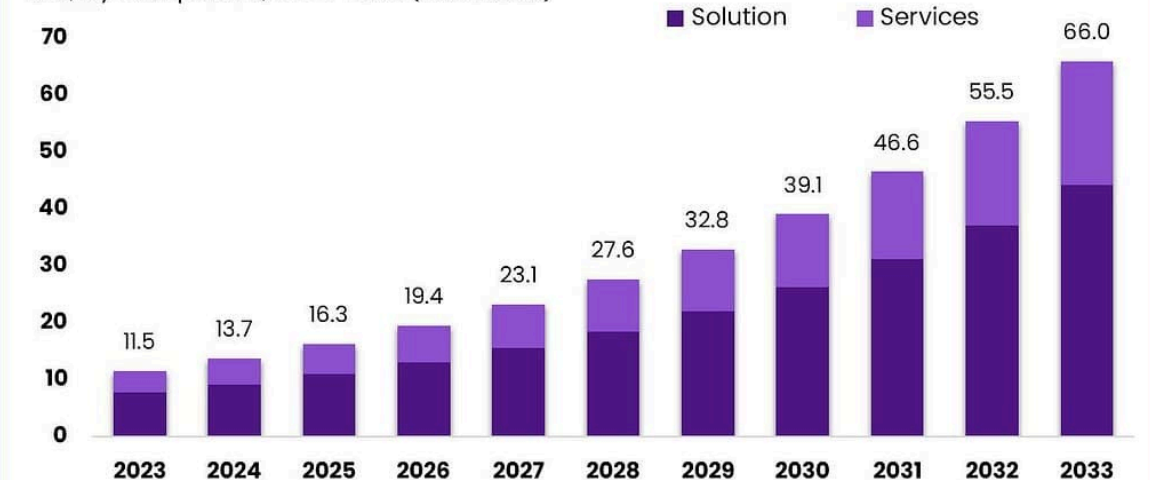
Total: 65 m / 100

Pre requisites

- ① Linux - CLI, basics
- ② Networking - Basics
- ③ Any programming lang/tech
 - Java/JS preferred
- ④ Mindset: Admin, Learning/Exploring, ...

Global DevOps Market

Size, by Component, 2024-2033 (USD Billion)



The Market will Grow
At the CAGR of:

19.1%

The Forecasted Market
Size for 2033 in USD:

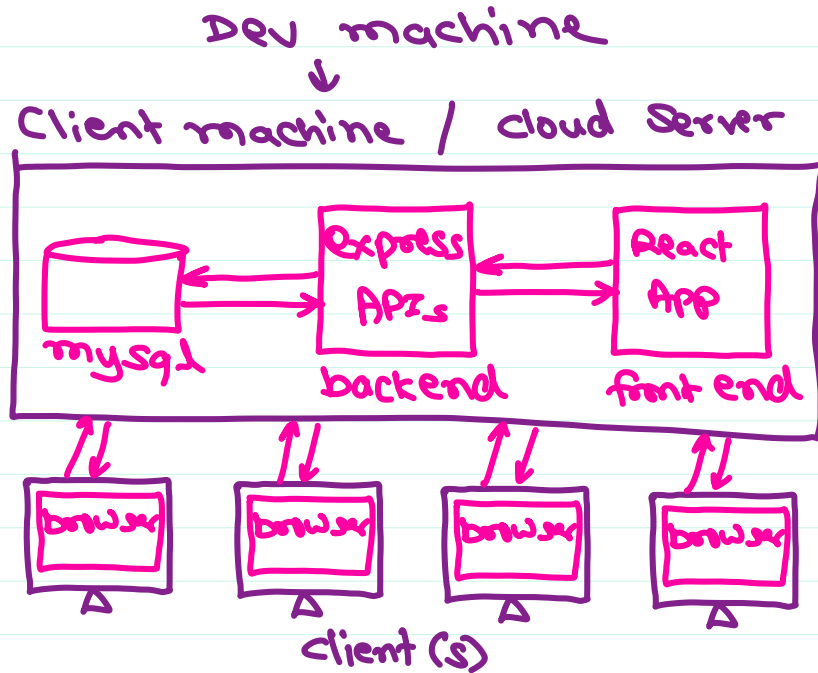
\$66.0B

market.us
ONE STOP SHOP FOR THE REPORTS

Application Deployment

$appn = code + runtime + libs + conf + os$

① multiple apps run on single machine



✓ cost effective

✗ SPOF

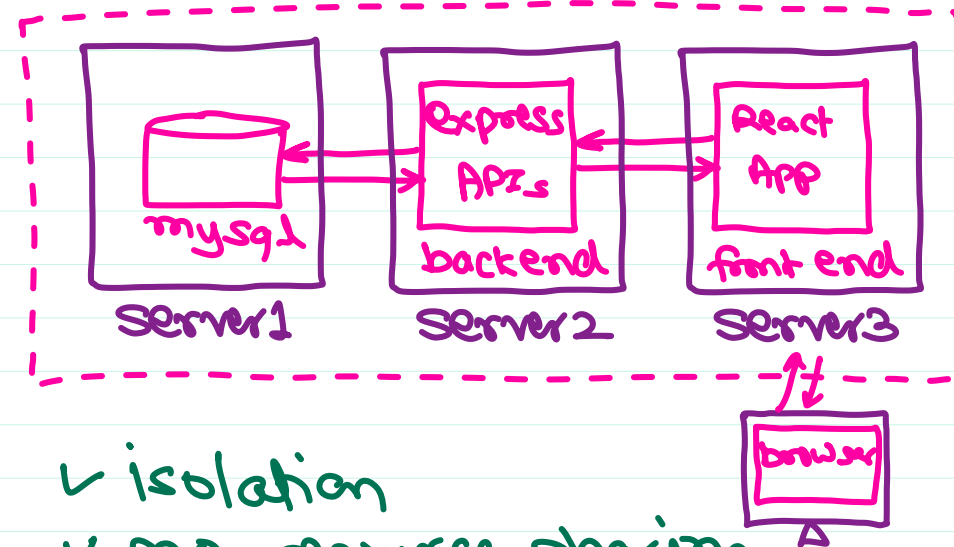
✗ resource sharing

✗ works on my machine

② multiple apps run on multiple machines

* Capital expenditure

* operational expenditure



✓ isolation

✓ no resource sharing

✗ Cost increases

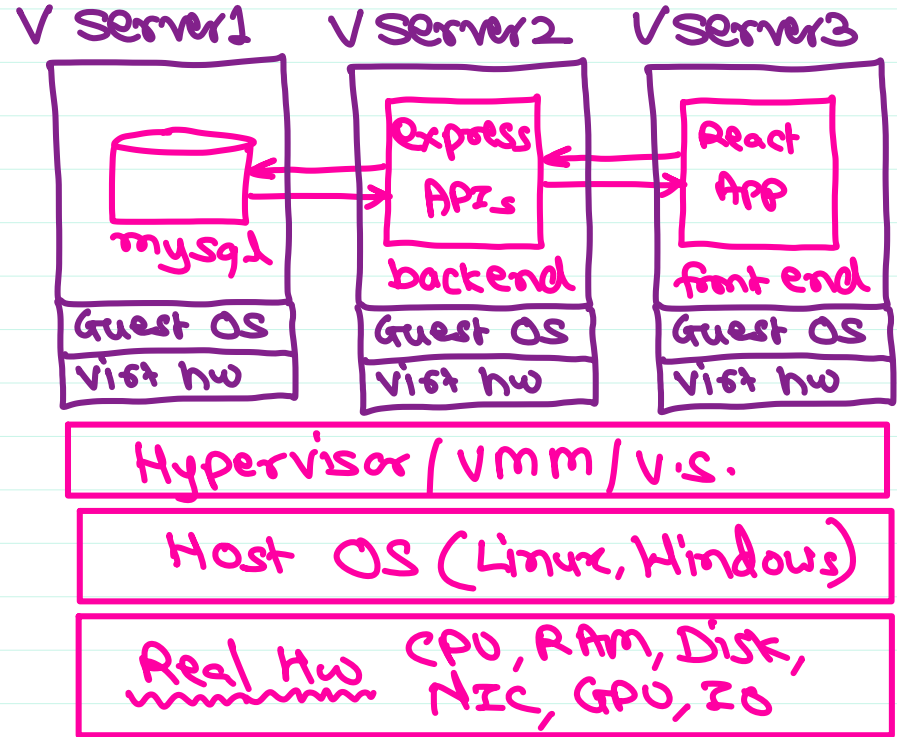
- separate machines ↑

- operational cost ↑

Virtualisation → vCPU, vGPU, vMemory, vIO, vDisk, vNIC, ...

③ hardware virtualization - Type II - Hypervisor

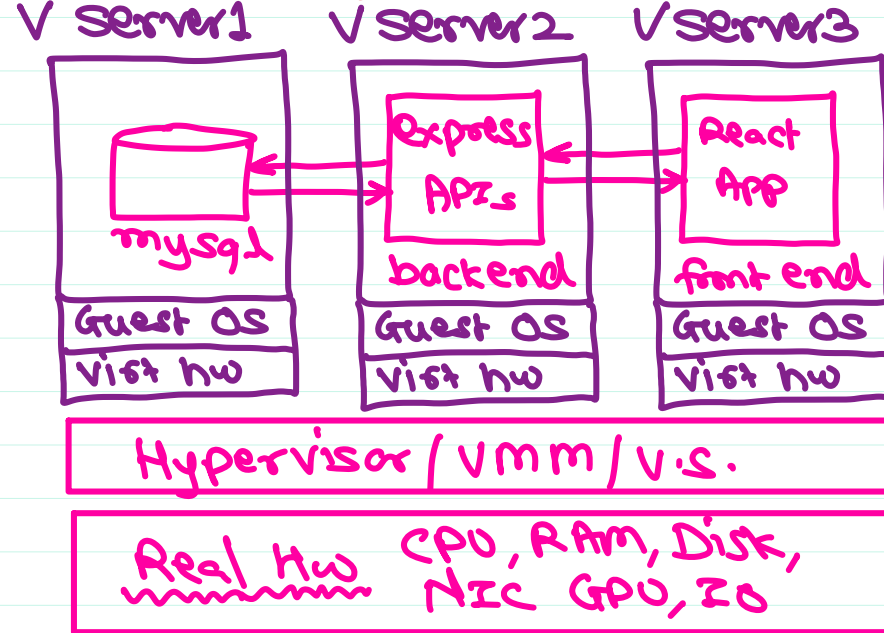
* Capital expenditure ↓



* advantages

- lower cost
- isolation
- backup
- deployment

④ hardware virtualization - Type I - Hypervisor



Vmware ESXi

* Data centers

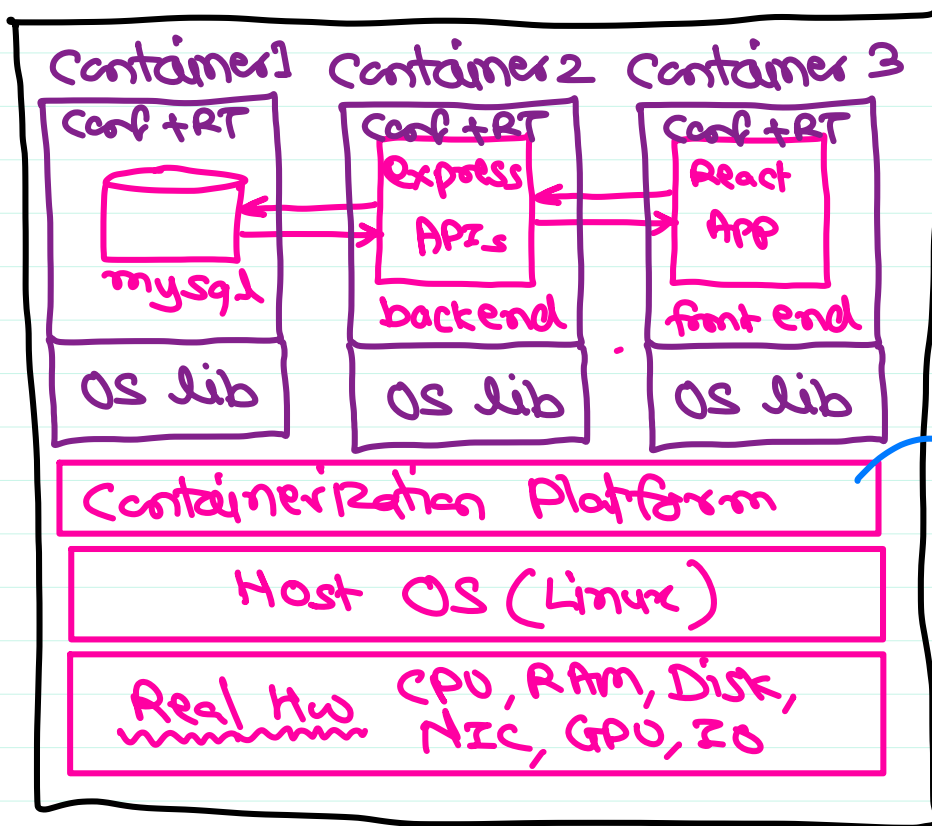
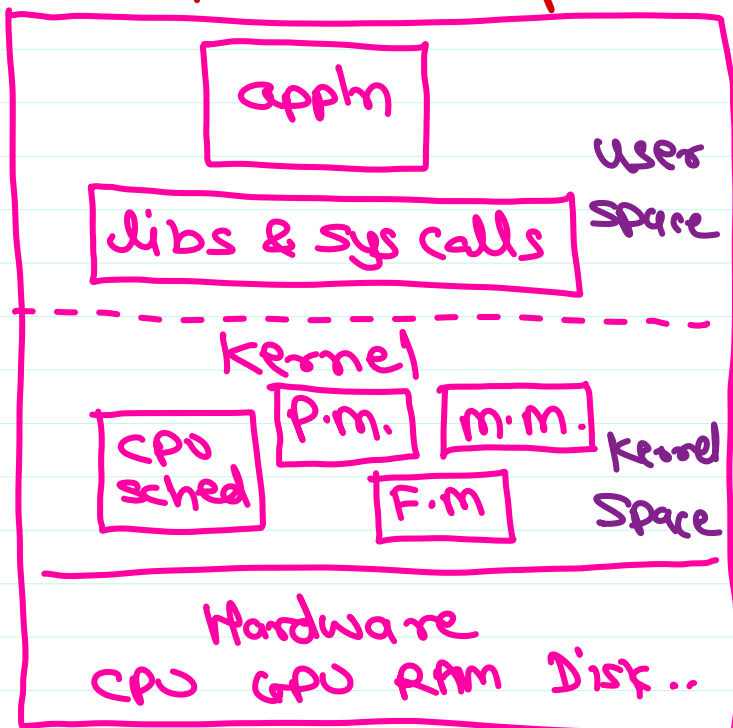
* limitations

- heavy OS
- boot time
- resource usage (mem, disk, cpu)
- OS duplication
- OS mgmt & licensing

Containerisation - OS virtualization

⑤ OS user space
vs kernel space

⑥ Containerization



vm	Container
Full OS	Shared Kernel
Heavy	Light weight
Slow boot	Fast startup
GBs	MBs
Hw virt.	OS isolation

Lxc
docker
podman
CRI-O

appm = code + dependencies + config + runtime + OS dep.

Container → appm exec environment

Docker

- Open-source containerization platform.
- Enables
 - App deployment
 - App isolation
 - App scaling
 - Security
 - Efficient exec.

History

- ① Solomon Hykes
2008 - dotCloud - PaaS
- Use Linux Containers to solve "works on my machine".
- ② Complex LXC to Simplified Docker (2013).
Docker = Dock Worker who handle Cargo Containers.
Open Sourced to Apache.

* Docker Installation
- follow official docs.

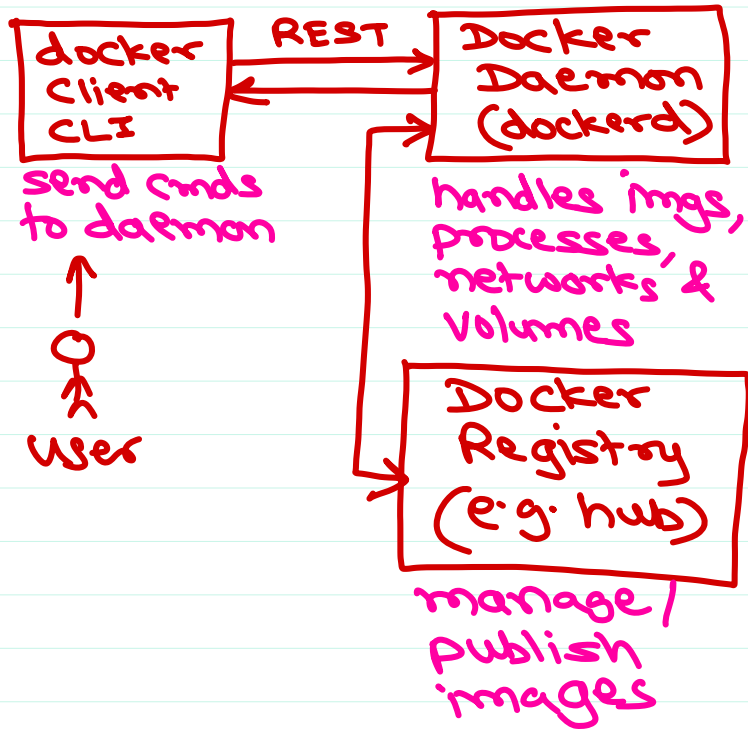
* Docker in action
> docker run hello-world
> docker run -p 8080:80 nginx

* Image → Container

- | | |
|----------------------------------|------------------------|
| ✓ read-only multi-layer template | ✓ an instance of image |
| ✓ Immutable | ✓ Thin writable layer |
| ✓ From DockerFile | ✓ From Image |
| ✓ Shared via reg. | ✓ Start, Stop, Destroy |

Docker

* Docker Architecture



Container Lifecycle

my-apache ← index.html

* PS [-a]

① create

cp index.html

② start

③ pause

④ unpause

⑤ stop

logs

⑥ rm [-f]

docker images

* images

① pull

② ~~run~~ = create + start

③ rmi

④ build from Docker file

⑤ push (?)





Thank You!

Nilesh Ghule

<nilesh@sunbeaminfo.com>